

Smart Room for Patients (IoT)

① THE PROBLEM - AND WHO IS FOR

Problem

Hospitalized patients need continuous monitoring while their room should remain comfortable and adapted for their condition.

Healthcare staff also need to be rapidly informed when abnormal patient conditions are detected

Target users

Hospitalized patients, nurses and doctors.

Proposed service

A smart hospital IoT system that monitors patient physiological parameters and room conditions, automatically controls the room environment, and generates alerts for healthcare staff when predefined abnormal conditions are detected

② WHAT'S SENSED - WHAT'S ACTED ON

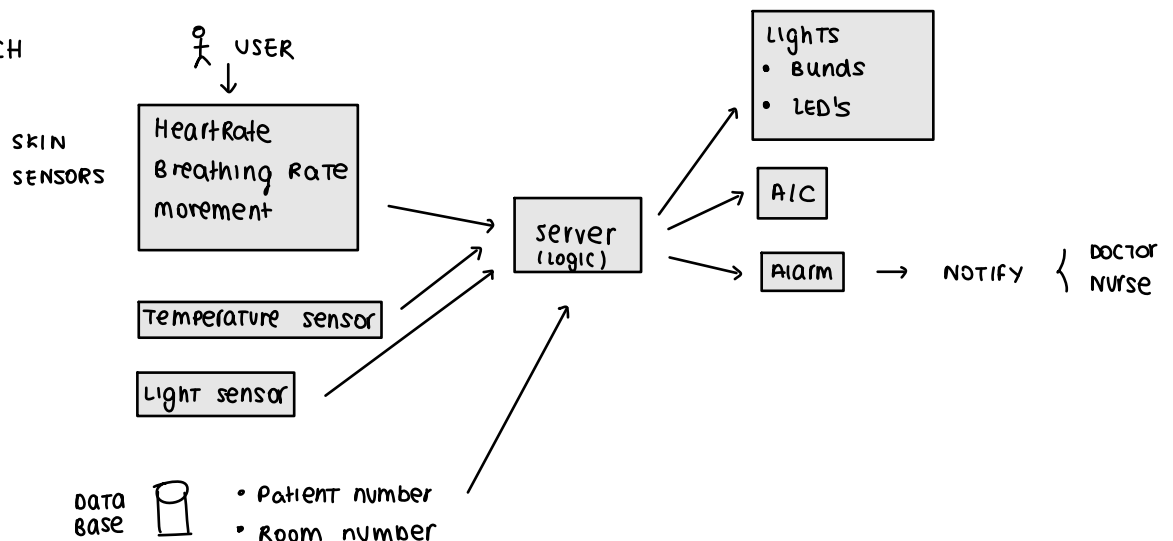
Sensed

- Heart Rate - Body sensor
- Breathing Rate - Body sensor
- Movement - Body sensor
- Room Temperature - Temperature sensor
- Ambient Light - Light sensor

Acted on

- LED / smart lights - adjust artificial light
- Monitored blinds - adjust natural light
- Air conditioner - adjust room temperature
- Alarm / Notification System - Alert healthcare staff
- Nurse / Doctor Notification - route the alert according to danger level

③ SYSTEM SKETCH



④ THREE TESTABLE REQUIREMENTS

R1 - PATIENT ALERT

When a monitored physiological parameter exceeds its configured threshold, the system shall generate an alert containing the patient ID, room number and detected abnormal parameter within 5s.

R2 - TEMPERATURE CONTROL

When measured room temperature doesn't match the desired temperature, the system should send an activation command for the A/C system

R3 - AUTOMATIC LIGHTING

When ambient light doesn't match the desired light threshold (based on the time of the day and patient parameters), the system shall adjust the lighting accordingly to the situation

⑤ OPEN QUESTIONS FOR AI.

- How should we manage the data base and the relation between patient ID, room number & sensor data?
- How should we implement the alert severity levels?
- How can we simulate data for the demo?
- How do we build the logical connections for the sensors, server and dashboard?
- What attributes does the json file have?